

FRC 4451 Badge 1 Pretest

Name: _____ Date: _____

Take this test in your first week, before any lesson, and again after the capstone.

Work on paper without notes or internet. Answer what you know. You may write “not sure,” leave a question blank, or show part of your reasoning. The mentor sets the session length and can clarify unfamiliar instructions.

1. The Shell

You open a terminal in your home folder. Write the commands, in order, that move into a folder named **robot**, list what is in it, and print the full path of the folder you are in. Specify if this works for PowerShell and/or Unix.

2. Networking

A laptop has address **192.168.1.20** and another device has **10.44.51.2**. Both use subnet mask **255.255.255.0** and no router connects their networks. Are they on the same subnet? Explain. Then name the one command you would run to find out whether a device answers on the network.

3. Git Terminology

In one sentence each, what is a **commit**, a **branch** and a **remote**?

4. Git Sequencing

Put these five steps in the order you would do them to get a change onto GitHub without touching **main**:

```
git push -u origin feature
git add .
edit a file
git commit -m "Describe the change"
git switch -c feature
```

5. Predict the Output

What does this print, one value per line?

```
int a = 7;
int b = 2;
System.out.println(a / b);
System.out.println(a % b);
System.out.println(a / 2.0);
```

6. Conditions and Boolean logic

The first snippet has one problem. Say what it is.

```
boolean enabled = false;
if (enabled = true) {
    System.out.println("running");
}
```

Then use De Morgan's law to rewrite this expression so the negation applies to each variable instead of the parentheses. The new expression must have the same result for every input.

```
!(enabled && connected)
```

7. Write a Loop

Write a loop that prints the numbers 10 down to 1, one per line.

[illegible]

8. Write a Method

Assume the method belongs inside a class. Write a **public static** method named **clamp** that takes a **double value**, a **double min** and a **double max**, and returns **value** limited to that range: **min** if it is below, **max** if it is above, otherwise **value** unchanged.

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9. Objects

The class below belongs to **Motor.java**. The four statements beneath it run inside another class's **main** method. What does the print statement output, and why?

```
public class Motor {  
    private double speed = 0;  
  
    public void set(double s) {  
        speed = s;  
    }  
  
    public double get() {  
        return speed;  
    }  
}  
  
Motor m = new Motor();  
Motor n = m;  
n.set(0.5);  
System.out.println(m.get());
```

10. Object Oriented Programming (OOP)

A program declares an interface with two implementations: one that talks to real hardware and one that pretends to, for testing in simulation. Name the Object-Oriented-Programming Principles applied. In two or three sentences, what does that arrangement make possible?
